| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 13(b) | 3780 | **2** | **M1** for $\frac{6^3}{1120}=\frac{9^3}{V}$ or better or $1120 \times \left(\frac{9}{6}\right)^3$ oe<br>or for $\left(\frac{9}{6}\times\sqrt[3]{1120}\right)^3$ |
| 14 | $(x-2)(5-a)$ final answer | **2** | **M1** for $5(x-2)-a(x-2)$<br>or for $x(5-a)-2(5-a)$<br>or $-5(2-x)+a(2-x)$<br>or $2(a-5)-x(a-5)$<br><br>or for correct answer seen then spoilt |
| 15 | 45 | **2** | **M1** for $\frac{360}{180-172}$ oe or $\frac{(n-2)180}{n}=172$ oe |
| 16(a) | 0.3 oe | **1** | |
| 16(b) | 0.69 oe | **3** | **M2** for $0.7 \times 0.9 + (\textit{their}\ 0.3) \times 0.2$ oe<br><br>or **M1** for $0.7 \times 0.9$ oe or $(\textit{their}\ 0.3) \times 0.2$ oe |
| 17(a) | 446 | **3** | **B2** for answer 46<br>or **M2** for $400+\frac{400\times 2.3\times 5}{100}$ oe<br>or **M1** for $\frac{400\times 2.3[\times 5]}{100}$ oe |
| 17(b) | 20 nfww | **3** | **M2** for $x + 1.1x + 1.1^2x = [100 -]\ 33.80$ oe<br><br>or **M1** for $\left(1+\frac{10}{100}\right)^2[x]$ oe seen<br>or for one correctly evaluated trial |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 17(c) | 742.97 to 742.99 | **4** | **B3** for 1.02[0…] or interest rate = 2[.0…][%]<br>OR<br>**M3** for $500\times\left(\sqrt[17]{\frac{700.13}{500}}\right)^{20}$ oe<br>or for $700.13\times\left(\sqrt[17]{\frac{700.13}{500}}\right)^{3}$ oe<br>or **M2** for $\sqrt[17]{\frac{700.13}{500}}$ oe<br>OR<br>**M1** for $500(\ldots)^{17} = 700.13$ oe<br>**M1** dep on previous M1 for<br>$500\times\left(1+\frac{their\ r}{100}\right)^{20}$ or $700.13\times\left(1+\frac{their\ r}{100}\right)^{3}$ |
| 18 | [$x$ =] 29<br>[$y$ =] 52<br>[$z$ =] 107 | **4** | **B1** for 29<br>**B1** for 52<br>**B2FT** for 107 or for $z$ = *their* $x$ + *their* $y$ + 26<br>or **B1** for $\angle EAC = 73$ or $\angle EBC = 73$<br>or **B1FT** for $\angle EAC = 180 - (\textit{their } x + \textit{their } y + 26)$ |
| 19(a) | –2 | **1** | |
| 19(b) | 11, 5, 1 | **2** | **B1** for 2 correct listed on answer line |
| 19(c) | –5 | **1** | |
| 19(d) | 8 | **2** | **M1** for [$x$ =] h(3) or [$x$ =] $2^3$ |
| 20(a) | 8.09 or 8.094… | **2** | **M1** for $\tan 34 = \frac{AB}{12}$ oe |
| 20(b) | 98.3 or 98.28 to 98.31 | **3** | **M1** for $\frac{1}{2}\times 10\times 12\sin 56$ oe<br>**M1** for $\frac{1}{2}\times 12\times$ *their* **(a)** oe |